

BUS123 · MATH-M08 · L01

Simple Interest & Discounting

Pre-Reading Guide · Meridian Advisory Group

Read this before class. Estimated time: 20 minutes. Simulated data for instructional purposes — Meridian Advisory Group is a fictional company.

1. Connecting to What You Already Know

In MATH-M07, you worked with percentages and rates — calculating markups, markdowns, and proportional changes. You learned that a rate is a ratio that describes change relative to a base amount. Today we put that skill to work in a new direction: using a rate to describe how money changes *over time*.

Until now, a 6% rate was a label — it described a discount or a price change at a single moment. Starting today, a 6% rate is an engine. It tells us how fast money grows when it sits in an account or a loan, and it tells us how much we need to 'shrink' a future promise to find out what it's worth right now. That shrinking process is called **discounting**, and it is the single most important calculation in business finance.

2. Why a Dollar Today Is Worth More Than a Dollar Tomorrow

The Time Value of Money (TVM) is the idea that money available today is worth more than the same amount of money promised in the future. This sounds obvious once you hear it, but understanding *why* it is true — and how to measure it — is what separates financial thinking from ordinary arithmetic.

Three forces explain why future money is worth less:

Opportunity Cost. Money available today can be invested immediately. A dollar sitting in your hand can earn a return. A dollar promised next year cannot earn anything until it arrives. Every day of waiting is a day of foregone earnings.

Inflation. Prices tend to rise over time. The same \$10,500 you receive next year will buy slightly less than \$10,500 buys today. Your nominal amount stays the same, but your purchasing power has eroded.

Risk. Future promises can fail. A payment guaranteed today carries no uncertainty. A payment promised in three years might not materialize — the borrower could default, the contract could be voided, or circumstances could change.

The So What: When a financial advisor at Meridian Advisory Group evaluates a client's future payment stream, they don't just add up the dollar amounts. They ask: what are these future dollars worth *today*, given the client's opportunity cost and the risk of waiting? That calculation — discounting — is what you will build in today's activity.

3. Simple Interest — Moving Money Forward in Time

Simple interest calculates the interest earned on a principal amount using a flat annual rate applied to the original principal only — interest does not earn interest (that is compound interest, covered next module). Simple interest is common in short-term loans, notes payable, and bridge financing.

The Formula:

$$I = P \times R \times T$$

Variable	Stands For	Notes
I	Interest earned (\$)	The dollar amount of interest generated
P	Principal (\$)	The original amount borrowed or invested
R	Annual interest rate	Always enter as a decimal: 6% = 0.06
T	Time in years	Must be in years — convert months ÷ 12, days ÷ 365

■ The T Trap — Never Use Days or Months Directly

The most common error on simple interest problems is plugging in time without converting to years first. A 90-day loan does NOT use $T = 90$. It uses $T = 90 \div 365 = 0.247$. A 6-month loan uses $T = 6 \div 12 = 0.5$. If you forget to convert, your interest will be 90 or 12 times too large — a result that should immediately signal something is wrong.

Worked Example — Meridian Advisory Group:

A Meridian client invests \$8,000 in a short-term corporate note at 5.5% simple interest for 9 months. What interest is earned, and what does the client receive at maturity?

Step 1: Convert time to years:

$$T = 9 \div 12 = 0.75 \text{ years}$$

Step 2: Calculate interest:

$$I = 8,000 \times 0.055 \times 0.75 = \$330.00$$

Step 3: Calculate Future Value:

$$FV = 8,000 + 330 = \$8,330.00$$

In Excel, Future Value is one formula:

$$=P*(1+R*T) \rightarrow =8000*(1+0.055*0.75) = \$8,330.00$$

Key insight: The term $(1 + R \times T)$ is the **growth factor**. It tells you how many times larger your money becomes. At 5.5% for 9 months, every dollar grows to \$1.04125. Multiply by the principal and you have the Future Value.

4. Discounting — Running the Math Backward

Future Value asks: *what will this grow to?* Present Value asks the reverse: *what do I need today to end up with a specific future amount?* — or equivalently, *what is a future promise worth right now?*

To get Present Value, we simply rearrange the Future Value formula. Instead of multiplying by the growth factor, we divide by it:

Direction	Question	Formula
Forward →	What will P grow to?	$FV = P \times (1 + R \times T)$
← Backward	What is FV worth today?	$PV = FV \div (1 + R \times T)$

The term $(1 + R \times T)$ in the denominator is the discount factor:

$$\text{Discount Factor} = 1 + R \times T$$

The discount factor is always greater than 1 (as long as R and T are positive), so dividing by it always produces a Present Value *smaller* than the Future Value. The higher the rate or the longer the wait, the larger the discount factor — and the smaller the present value. This is the mathematical proof of TVM: future cash is always discounted, never inflated.

Worked Example — Meridian Advisory Group:

A Meridian corporate client is owed \$25,000 in 18 months from a contract receivable. They want to know what that future payment is worth today, using a 7% discount rate.

Step 1: Convert time to years:

$$T = 18 \div 12 = 1.5 \text{ years}$$

Step 2: Calculate discount factor:

$$DF = 1 + (0.07 \times 1.5) = 1.105$$

Step 3: Calculate Present Value:

$$PV = 25,000 \div 1.105 = \$22,624.43$$

In Excel:

$$=FV/(1+R \times T) \rightarrow =25000/(1+0.07 \times 1.5) = \$22,624.43$$

The gap tells the story: \$25,000 – \$22,624 = \$2,376. That \$2,376 is the cost of waiting 18 months at 7%. It represents the combined effect of opportunity cost and risk. This gap — between future face value and present value — is what the entire field of discounted cash flow (DCF) analysis is built on.

5. Formula Reference

Formula	Algebraic Form	Excel Form	Use When
Simple Interest	$I = P \times R \times T$	$=C2 \times D2 \times E2$	Finding interest earned on a short-term note
Future Value	$FV = P \times (1 + R \times T)$	$=C2 \times (1 + D2 \times E2)$	Finding the total maturity value
Present Value	$PV = FV \div (1 + R \times T)$	$=C2 / (1 + D2 \times E2)$	Discounting a future payment to today
Discount Factor	$DF = 1 + R \times T$	$=1 + D2 \times E2$	Intermediate step in PV calculation

Time Conversion	$T = \text{months} \div 12 \text{ or } \text{days} \div 365$	$=A2/12 \text{ or } =A2/365$	Always convert before using in formulas
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6. Check Your Understanding

Answer these seven questions before class. The answer key appears on the final page. Show your work on questions 2–5.

1. (Conceptual)

In one or two sentences, explain why a financial advisor would tell a client that \$15,000 promised in two years is *not* worth \$15,000 today. Do not use a formula — explain the economic logic.

2. (Calculation)

A Meridian client borrows \$10,000 at 6% simple interest for 2 years. Calculate (a) the interest earned and (b) the future value at maturity.

3. (Calculation)

A 180-day commercial note has a principal of \$7,500 and an annual rate of 4.8%. Calculate the interest earned using the exact interest method (365-day year).

4. (Calculation)

A Meridian client will receive \$18,000 in 2 years. Using a 7% discount rate, calculate the present value of that payment.

5. (Calculation)

Using a 6% discount rate, calculate the present value of each payment, then find the total present value of this two-payment stream: • Payment A: \$12,000 due in 1 year • Payment B: \$20,000 due in 3 years

6. (Conceptual)

A student calculates the present value of a payment and gets an answer larger than the future value. Without doing the math, explain what error the student must have made.

7. (Analysis)

Suppose the discount rate in Question 4 increases from 7% to 12%. Without recalculating, predict whether the present value will be higher or lower than your answer to Question 4. Then calculate the new PV to verify your prediction.

7. Key Vocabulary

Time Value of Money (TVM)	The principle that a dollar today is worth more than a dollar in the future, because today's dollar can be invested to earn a return.
Principal (P)	The original amount of money borrowed, invested, or lent — before interest is added.
Simple Interest	Interest calculated only on the original principal. Interest does not earn interest.
Future Value (FV)	The value a present sum of money will reach at a specified future date, given a specific interest rate and time period.

Present Value (PV)	The current worth of a future sum of money, calculated by discounting at a specific rate. Also called the discounted value.
Discount Rate	The interest rate used to convert future cash flows back to present value. It reflects the investor's required return — their opportunity cost plus a risk premium.
Discount Factor	The denominator in the PV formula: $(1 + R \times T)$. A number always greater than 1 that shrinks a future value down to its present equivalent.
Discounted Cash Flow (DCF)	A valuation method that finds the present value of a stream of future cash flows by discounting each one back to today and summing the results.
Opportunity Cost	The return foregone by choosing one option over the next best alternative. In TVM, it is why money today is more valuable — you could be earning a return on it right now.
Exact Interest	Simple interest calculated using a 365-day year in the denominator for time conversion.

Answer Key — Check Your Understanding

1. (Conceptual)

Three forces reduce the value of future money: (1) opportunity cost — the \$15,000 today could be invested to earn a return over the two years; (2) inflation — prices will likely rise, so \$15,000 will buy less in two years; (3) risk — the future payment might not be received. Any two of these forces, explained clearly, is a complete answer.

2. (Calculation)

(a) $I = 10,000 \times 0.06 \times 2 = \$1,200.00$ (b) $FV = 10,000 + 1,200 = \$11,200.00$ [or: $=10000 \times (1+0.06 \times 2) = \$11,200.00$]

3. (Calculation)

$T = 180 \div 365 = 0.4932$ years $I = 7,500 \times 0.048 \times 0.4932 = \177.53

4. (Calculation)

$DF = 1 + (0.07 \times 2) = 1.14$ $PV = 18,000 \div 1.14 = \$15,789.47$

5. (Calculation)

Payment A: $PV = 12,000 \div (1 + 0.06 \times 1) = 12,000 \div 1.06 = \$11,320.75$ Payment B: $PV = 20,000 \div (1 + 0.06 \times 3) = 20,000 \div 1.18 = \$16,949.15$ Total $PV = \$11,320.75 + \$16,949.15 = \$28,269.90$

6. (Conceptual)

The Present Value must always be less than the Future Value (because the discount factor is always greater than 1, making the division produce a smaller result). If $PV > FV$, the student likely multiplied instead of divided by the discount factor — running the formula in the wrong direction.

7. (Analysis)

Prediction: PV will be lower. A higher discount rate means a larger denominator, which produces a smaller present value. This reflects higher perceived risk — the investor demands a greater return for waiting. Verification: $DF = 1 + (0.12 \times 2) = 1.24$ New $PV = 18,000 \div 1.24 = \$14,516.13$ (vs. $\$15,789.47$ at 7% — lower, as predicted)

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